Unifying (Herbrand) functional interpretation of (nonstandard) arithmetic

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My previous talk

Extracting Computer Programs from Nonstandard Proofs

An Agda implementation of

> *Benno van den Berg, Eyvind Briseid, and Pavol Safarik, A functional interpretation for nonstandard arithmetic, Annals of Pure and Applied Logic 163 (2012), no. 12, 1962–1994.*

- We reformulate the Herbrand functional interpretations in a way that is suitable for a type-theoretic development.
- Then we embed the nonstandard arithmetic within Agda and implement the term extraction algorithm behind the functional interpretations.

This talk

We unify the Herbrand functional interpretations of nonstandard arithmetic following Oliva[1] by introducing a parametrised functional interpretation such that

- We have a parametrised soundness proof.
- The Herbrand functional interpretations of nonstandard arithmetic are obtained via its instantiations.
- The usual functional interpretations of uniform Heyting arithmetic[2] can also be obtained via its instantiations.

[1] Paulo Oliva, *Unifying functional interpretations*, Notre Dame J. Formal Logic 47 (2006), no. 2, 263–290.

[2] Ulrich Berger, *Uniform Heyting arithmetic*, Annals of Pure and Applied Logic 133 (2005), no. 1, 125–148.

Heyting arithmetic with finite types HA^ω

Term language T:

Simply typed lambda calculus (or SKI) + natural numbers and recursor

Logic language:

Intuitionistic logic + arithmetic axioms (incl. the induction axiom)

- Equality of natural numbers only ($I\text{-}HA^\omega$)
  so that its Dialectica interpretation is sound
- Can be embedded as inductive datatypes within dependent type theory

A nonstandard arithmetic

Term language T^*: T + finite sequences σ^*

to simulate finite sets for formulating the nonstandard axioms

$\mathsf{HA}^{\omega *} :\equiv \mathsf{HA}^\omega$ + axioms for finite sequences

$\mathsf{HA}^{\omega *}_{\mathsf{st}} :\equiv \mathsf{HA}^{\omega *}$ + st predicate + axioms for st + external induction principle

$$\Phi(0) \land \forall^{\mathrm{st}} n\ (\Phi(n) \to \Phi(\mathsf{s}n)) \ \to \ \forall^{\mathrm{st}} n\ \Phi(n)$$

We add $\forall^{\mathrm{st}}, \exists^{\mathrm{st}}$ and axioms $\forall^{\mathrm{st}} x A \leftrightarrow \forall x(\mathsf{st}(x) \to A),\ \exists^{\mathrm{st}} x A \leftrightarrow \exists x(\mathsf{st}(x) \land A)$

System $\mathsf{H} :\equiv \mathsf{HA}^{\omega *}_{\mathsf{st}}$ + 5 nonstandard axioms (characterisation of Dialectica)

Herbrand Dialectica interpretation

Idea: Each formula $\Phi(\underline{a})$ in $\mathsf{HA}^{\omega *}_{\mathrm{st}}$ is interpreted as $\exists^{\mathrm{st}}\underline{x}\forall^{\mathrm{st}}\underline{y}\,\varphi_{\mathsf{D}_{\mathrm{st}}}(\underline{a},\underline{x},\underline{y})$ where $\underline{x}$ is a finite sequence of potential realisers, and $\varphi_{\mathsf{D}_{\mathrm{st}}}(\underline{a},\underline{x},\underline{y})$ is internal.

In van den Berg *et al.*, it is (informally) defined as follows

(i) $\varphi(\underline{a})^{D_{\mathrm{st}}} :\equiv \varphi_{D_{\mathrm{st}}}(\underline{a}) :\equiv \varphi(\underline{a})$ for internal atomic formulas $\varphi(\underline{a})$,
(ii) $\mathrm{st}^{\sigma}(u^{\sigma})^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}x^{\sigma *}\ u \in_{\sigma} x$.

Let $\Phi(\underline{a})^{D_{\mathrm{st}}} \equiv \exists^{\mathrm{st}}\underline{x}\ \forall^{\mathrm{st}}\underline{y}\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},\underline{a})$ and $\Psi(\underline{b})^{D_{\mathrm{st}}} \equiv \exists^{\mathrm{st}}\underline{u}\ \forall^{\mathrm{st}}\underline{v}\ \psi_{D_{\mathrm{st}}}(\underline{u},\underline{v},\underline{b})$. Then

(iii) $(\Phi(\underline{a}) \wedge \Psi(\underline{b}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{x},\underline{u}\ \forall^{\mathrm{st}}\underline{y},\underline{v}\ (\varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},\underline{a}) \wedge \psi_{D_{\mathrm{st}}}(\underline{u},\underline{v},\underline{b}))$,
(iv) $(\Phi(\underline{a}) \vee \Psi(\underline{b}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{x},\underline{u}\ \forall^{\mathrm{st}}\underline{y},\underline{v}\ (\varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},\underline{a}) \vee \psi_{D_{\mathrm{st}}}(\underline{u},\underline{v},\underline{b}))$,
(v) $(\Phi(\underline{a}) \to \Psi(\underline{b}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{U},\underline{Y}\ \forall^{\mathrm{st}}\underline{x},\underline{v}\ (\forall \underline{y} \in \underline{Y}[\underline{x},\underline{v}]\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},\underline{a}) \to \psi_{D_{\mathrm{st}}}(\underline{U}[\underline{x}],\underline{v},\underline{b}))$.

Let $\Phi(z,\underline{a})^{D_{\mathrm{st}}} \equiv \exists^{\mathrm{st}}\underline{x}\ \forall^{\mathrm{st}}\underline{y}\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},z,\underline{a})$, with the free variable z not occurring among the $\underline{a}$. Then

(vi) $(\forall z\ \Phi(z,\underline{a}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{x}\ \forall^{\mathrm{st}}\underline{y}\ \forall z\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y},z,\underline{a})$,
(vii) $(\exists z\ \Phi(z,\underline{a}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{x}\ \forall^{\mathrm{st}}\underline{y}\ \exists z\ \forall \underline{y'} \in \underline{y}\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y'},z,\underline{a})$,
(viii) $(\forall^{\mathrm{st}}z\ \Phi(z,\underline{a}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{X}\ \forall^{\mathrm{st}}z,\underline{y}\ \varphi_{D_{\mathrm{st}}}(\underline{X}[z],\underline{y},z,\underline{a})$,
(ix) $(\exists^{\mathrm{st}}z\ \Phi(z,\underline{a}))^{D_{\mathrm{st}}} :\equiv \exists^{\mathrm{st}}\underline{x},z\ \forall^{\mathrm{st}}\underline{y}\ \exists z' \in z\ \forall \underline{y'} \in \underline{y}\ \varphi_{D_{\mathrm{st}}}(\underline{x},\underline{y'},z',\underline{a})$.

Types of realisers and counterexamples

For a formal (type-theoretic) development[3], we calculate the types $\mathsf{d}^+\Phi$ of (actual) realisers and $\mathsf{d}^-\Phi$ of counterexamples for formula Φ:

$$\begin{aligned}
\mathsf{d}^+(a =_\sigma b) &:\equiv \mathbb{1} \\
\mathsf{d}^+(\mathsf{st}^\sigma(t)) &:\equiv \sigma \\
\mathsf{d}^+(A \wedge B) &:\equiv \mathsf{d}^+A \times \mathsf{d}^+B \\
\mathsf{d}^+(A \vee B) &:\equiv \mathsf{d}^+A \times \mathsf{d}^+B \\
\mathsf{d}^+(A \to B) &:\equiv ((\mathsf{d}^+A)^* \to (\mathsf{d}^+B)^*) \times ((\mathsf{d}^+A)^* \to \mathsf{d}^-B \to (\mathsf{d}^-A)^*) \\
\mathsf{d}^+(\forall x^\sigma A) &:\equiv \mathsf{d}^+A \\
\mathsf{d}^+(\exists x^\sigma A) &:\equiv \mathsf{d}^+A \\
\mathsf{d}^+(\forall^{\mathrm{st}} x^\sigma A) &:\equiv \sigma \to (\mathsf{d}^+A)^* \\
\mathsf{d}^+(\exists^{\mathrm{st}} x^\sigma A) &:\equiv \sigma \times \mathsf{d}^+A
\end{aligned}$$

$$\begin{aligned}
\mathsf{d}^-(a =_\sigma b) &:\equiv \mathbb{1} \\
\mathsf{d}^-(\mathsf{st}(t)) &:\equiv \mathbb{1} \\
\mathsf{d}^-(A \wedge B) &:\equiv \mathsf{d}^-A \times \mathsf{d}^-B \\
\mathsf{d}^-(A \vee B) &:\equiv \mathsf{d}^-A \times \mathsf{d}^-B \\
\mathsf{d}^-(A \to B) &:\equiv (\mathsf{d}^+A)^* \times \mathsf{d}^-B \\
\mathsf{d}^-(\forall x^\sigma A) &:\equiv \mathsf{d}^-A \\
\mathsf{d}^-(\exists x^\sigma A) &:\equiv (\mathsf{d}^-A)^* \\
\mathsf{d}^-(\forall^{\mathrm{st}} x^\sigma A) &:\equiv \sigma \times \mathsf{d}^-A \\
\mathsf{d}^-(\exists^{\mathrm{st}} x^\sigma A) &:\equiv (\mathsf{d}^-A)^*
\end{aligned}$$

- Compare to the original Dialectica interpretation ($\mathsf{st}, \forall^{\mathrm{st}}, \exists^{\mathrm{st}}, *$)
- Variables quantified by $\forall, \exists$ have no computational contents

[3] Helmut Schwichtenberg and Stanley S. Wainer, *Proofs and computations*, Cambridge University Press, 2011

Our formulation of the Herbrand Dialectica interpretation

For every formula Φ and terms $r : (\mathsf{d}^+\Phi)^*$ and $u : \mathsf{d}^-\Phi$, we define an internal formula $\Phi_{\mathsf{D_{st}}}(r,u)$ by induction on Φ:

$$\begin{aligned}
(a =_\sigma b)_{\mathsf{D_{st}}}(r,u) &:\equiv a =_\sigma b\\
(\mathsf{st}^\sigma(t))_{\mathsf{D_{st}}}(r,u) &:\equiv t \in_\sigma r\\
(A \wedge B)_{\mathsf{D_{st}}}(r,(u,v)) &:\equiv A_{\mathsf{D_{st}}}(r_1,u) \wedge B_{\mathsf{D_{st}}}(r_2,v)\\
(A \vee B)_{\mathsf{D_{st}}}(r,(u,v)) &:\equiv A_{\mathsf{D_{st}}}(r_1,u) \vee B_{\mathsf{D_{st}}}(r_2,v)\\
(A \to B)_{\mathsf{D_{st}}}(W,(r,v)) &:\equiv \forall u{\in}W_2[r,v]\, A_{\mathsf{D_{st}}}(r,u) \to B_{\mathsf{D_{st}}}(W_1[r],u)\\
(\forall z^\sigma \Phi(z))_{\mathsf{D_{st}}}(r,u) &:\equiv \forall z^\sigma (\Phi(z))_{\mathsf{D_{st}}}(r,u)\\
(\exists z^\sigma \Phi(z))_{\mathsf{D_{st}}}(r,u) &:\equiv \exists z^\sigma \forall v{\in}u\, (\Phi(z))_{\mathsf{D_{st}}}(r,v)\\
(\forall^{\mathrm{st}} z^\sigma \Phi(z))_{\mathsf{D_{st}}}(R,(a,u)) &:\equiv (\Phi(a))_{\mathsf{D_{st}}}(R[a],u)\\
(\exists^{\mathrm{st}} z^\sigma \Phi(z))_{\mathsf{D_{st}}}(r,u) &:\equiv \exists z{\in}r_1 \forall v{\in}u\, (\Phi(z))_{\mathsf{D_{st}}}(r_2,v)
\end{aligned}$$

The Herbrand Dialectica interpretation $\Phi^{\mathsf{D_{st}}}$ of a formula Φ is defined by

$$\Phi^{\mathsf{D_{st}}} :\equiv \exists^{\mathrm{st}} x^{(\mathsf{d}^+\Phi)^*}\, \forall^{\mathrm{st}} y^{\mathsf{d}^-\Phi}\, \Phi_{\mathsf{D_{st}}}(x,y)$$

Soundness of the Herbrand Dialectica interpretation

Theorem. Let Φ be a formula of system H and let Δ_{int} be a set of internal formulas. If

$$\mathsf{H} + \Delta_{\mathrm{int}} \vdash \Phi$$

then from the proof we can extract a closed term $t : (\mathsf{d}^+\Phi)^*$ in T^* such that

$$\mathsf{HA}^{\omega *} + \Delta_{\mathrm{int}} \vdash \forall y^{\mathsf{d}^-\Phi} \Phi_{\mathsf{D}_{\mathrm{st}}}(t, y).$$

Proof. By induction on the length of the derivation.

System H within Agda

http://cj-xu.github.io/agda/nonstandard_dialectica/Index.html

- As 4 dependently typed, inductive definitions
  (can't directly use Agda terms because of the nonstandard axioms)
- Natural deduction system
  (rather than the axiomatic one in van den Berg *et al.*)
- Embed T^* into Agda
  (instead of implementing an evaluator/normaliser)

Extracting Agda programs from proofs in H

http://cj-xu.github.io/agda/nonstandard_dialectica/Index.html

- We define H as 4 dependently typed, inductive datatypes
- We implement only the term extraction algorithm behind the soundness proof of the Herbrand Dialectica interpretation.
- We removed noncomputational parts (of the unit type) to simplify the extracted T^*-terms.
- We embed T^* into Agda (instead of implementing an evaluator) to obtain Agda programs from the extracted T^*-terms.
- The complete soundness proof is too complicated to implement within intensional type theory (to be discussed later).

Difficulty of the implementation in intensional type theory

In intensional type theory, for arbitrary formula Φ, we have

$$\mathsf{d}^{+/-}(\Phi) = \mathsf{d}^{+/-}(\Phi[x := t])$$

only up to identity type (similar to case $\Pi(n, m{:}\mathbb{N}).\, n + m = m + n$).

Then, given $r : \mathsf{d}^{+}(\Phi)$ we have to transport it along the above equality/path to get an element of $\mathsf{d}^{+}(\Phi[x := t])$.

Since the above equality is (implicitly) used in many places of the soundness proof, we have to apply the transport function many times and also need a few lemmas about those transported terms.

This makes proving the soundness theorem very difficult and the resulting proof unreadable.

Herbrand realisability

We firstly work out the types $\tau(\Phi)$ of (acutal) realisers for formula Φ.
Then for each formula Φ and term $s : (\tau\Phi)^*$ we define $s \text{ hr } \Phi$

$$
\begin{array}{lcl@{\qquad}lcl}
\tau(a =_\sigma b) & :\equiv & \mathbb{1} & s \text{ hr } a = b & :\equiv & a = b \\
\tau(\mathsf{st}^\sigma(t)) & :\equiv & \sigma & s \text{ hr } \mathsf{st}(t) & :\equiv & t \in s \\
\tau(A \wedge B) & :\equiv & \tau A \times \tau B & s \text{ hr } (A \wedge B) & :\equiv & s^1 \text{ hr } A \ \wedge \ s^2 \text{ hr } B \\
\tau(A \vee B) & :\equiv & \tau A \times \tau B & s \text{ hr } (A \vee B) & :\equiv & s^1 \text{ hr } A \vee \ s^2 \ \text{ hr } B \\
\tau(A \to B) & :\equiv & (\tau A)^* \to (\tau B)^* & s \text{ hr } (A \to B) & :\equiv & \forall^{\mathrm{st}} u(u \text{ hr } A \to s[u] \text{ hr } B) \\
\tau(\forall x^\sigma A) & :\equiv & \tau A & s \text{ hr } \forall x A(x) & :\equiv & \forall x(s \text{ hr } A(x)) \\
\tau(\exists x^\sigma A) & :\equiv & \tau A & s \text{ hr } \exists x A(x) & :\equiv & \exists x(s \text{ hr } A(x)) \\
\tau(\forall^{\mathrm{st}} x^\sigma A) & :\equiv & \sigma \to (\tau A)^* & s \text{ hr } \forall^{\mathrm{st}} x A(x) & :\equiv & \forall^{\mathrm{st}} x(s[x] \text{ hr } A(x)) \\
\tau(\exists^{\mathrm{st}} x^\sigma A) & :\equiv & \sigma \times (\tau A)^* & s \text{ hr } \exists^{\mathrm{st}} x A(x) & :\equiv & \exists z \in s^1(s^2 \text{ hr } A(z))
\end{array}
$$

Similar to the situation of (standard) Dialectica and modified realisability, their Herbrand variants differ in the interpretation of implication.

First attempt to unify Herbrand functional interpretations

As in Oliva 2006, one can introduce an uninterpreted bounded universal quantifier $\forall x \sqsubset t\, A(x)$ where $x : \sigma$ is a variable and $t : \sigma^*$ is a term.

Then the parametrised formula interpretation $|A|^x_y$ is almost the same as the D_{st}-interpretation except the case of implication

$$|A \to B|^R_{s,u} \;:\equiv\; \forall v \sqsubset R^2[s,u]\, |A|^s_v \to |B|^{R^1[s]}_u.$$

If we take $\forall x \sqsubset t\, A(x)$ to be $\forall x \in t\, A(x)$, then we get the Herbrand Dialectica.

If we take $\forall x \sqsubset t\, A(x)$ to be $\forall^{\mathrm{st}} x A(x)$, then we get the Herbrand realisability (because $s \text{ hr } A \;\leftrightarrow\; \forall^{\mathrm{st}} u |A|^s_u$).

Parametrised formula interpretation I

We want a more general parametrised formula interpretation to obtain also the standard functional interpretations via its instantiations.

The interpreted system: $\mathsf{HA}^{\omega *}_{\mathsf{st}} \equiv \mathsf{HA}^{\omega *} + \mathsf{st}$

The verifying system: $\mathsf{HA}^{\circ} \equiv \mathsf{HA}^{\omega *} + \sigma^{\circ} + t \,\epsilon\, w + \forall x \sqsubset t\, A(x)$

- σ° behaves as the type of finite sequences, e.g.
  - 'singleton' $\sigma \to \sigma^{\circ}$
  - 'concatenation' $\sigma^{\circ} \times \sigma^{\circ} \to \sigma^{\circ}$
  - 'pairing' $\sigma^{\circ} \times \rho^{\circ} \to (\sigma \times \rho)^{\circ}$
  - 'projections' $(\sigma_0 \times \sigma_1)^{\circ} \to \sigma_i$
  - 'application' $(\sigma \to \rho^{\circ})^{\circ} \times \sigma^{\circ} \to \rho^{\circ}$
- $t \,\epsilon\, w$ behaves as the membership relation
  for $t : \sigma$ and $w : \sigma^{\circ}$
- $\forall x \sqsubset w\, A(x)$ behaves as a bounded, universal quantifier
  for $x : \sigma$ and $w : \sigma^{\circ}$

Parametrised formula interpretation II

Each formula Φ is associated with types $\tau^+\Phi$ and $\tau^-\Phi$:

$$\tau^+(a =_\sigma b) :\equiv \mathbb{1} \qquad \tau^-(a =_\sigma b) :\equiv \mathbb{1}$$
$$\tau^+(\mathsf{st}^\sigma(t)) :\equiv \sigma \qquad \tau^-(\mathsf{st}(t)) :\equiv \mathbb{1}$$
$$\tau^+(A \wedge B) :\equiv \tau^+A \times \tau^+B \qquad \tau^-(A \wedge B) :\equiv \tau^-A \times \tau^-B$$
$$\tau^+(A \vee B) :\equiv \tau^+A \times \tau^+B \qquad \tau^-(A \vee B) :\equiv \tau^-A \times \tau^-B$$
$$\tau^+(A \to B) :\equiv ((\tau^+A)^\circ \to (\tau^+B)^\circ) \times ((\tau^+A)^\circ \times \tau^-B \to (\tau^-A)^\circ) \qquad \tau^-(A \to B) :\equiv (\tau^+A)^\circ \times \tau^-B$$
$$\tau^+(\forall x^\sigma A) :\equiv \tau^+A \qquad \tau^-(\forall x^\sigma A) :\equiv \tau^-A$$
$$\tau^+(\exists x^\sigma A) :\equiv \tau^+A \qquad \tau^-(\exists x^\sigma A) :\equiv (\tau^-A)^\circ$$
$$\tau^+(\forall^{\mathrm{st}} x^\sigma A) :\equiv \sigma \to (\tau^+A)^\circ \qquad \tau^-(\forall^{\mathrm{st}} x^\sigma A) :\equiv \sigma \times \tau^-A$$
$$\tau^+(\exists^{\mathrm{st}} x^\sigma A) :\equiv \sigma \times \tau^+A \qquad \tau^-(\exists^{\mathrm{st}} x^\sigma A) :\equiv (\tau^-A)^\circ$$

For each formula Φ and terms $r : (\tau^+\Phi)^\circ$ and $u : \tau^-\Phi$, we define formula $|\Phi|^r_u$

$$|a =_\sigma b|^r_u \;:\equiv\; a =_\sigma b \qquad |\forall z^\sigma \Phi(z)|^r_u \;:\equiv\; \forall z^\sigma |\Phi(z)|^r_u$$
$$|\mathsf{st}^\sigma(t)|^r_u \;:\equiv\; t \,\epsilon\, r \qquad |\exists z^\sigma \Phi(z)|^r_u \;:\equiv\; \exists z^\sigma \forall v \,\epsilon\, u |\Phi(z)|^r_v$$
$$|A \wedge B|^r_u \;:\equiv\; |A|^{r^1}_{u_1} \wedge |B|^{r^2}_{u_2} \qquad |\forall^{\mathrm{st}} z^\sigma \Phi(z)|^R_{a,u} \;:\equiv\; |\Phi(a)|^{R[a]}_u$$
$$|A \vee B|^r_u \;:\equiv\; |A|^{r^1}_{u_1} \vee |B|^{r^2}_{u_2} \qquad |\exists^{\mathrm{st}} z^\sigma \Phi(z)|^r_u \;:\equiv\; \exists z \,\epsilon\, r^1 \forall v \,\epsilon\, u |\Phi(z)|^{r^2}_v$$
$$|A \to B|^R_u \;:\equiv\; \forall v \sqsubset R^2[u] \, |A|^{u_1}_v \to |B|^{R^1[u_1]}_{u_2}$$

Parametrised formula interpretation $\mathsf{P}_{\mathsf{st}}(\Phi) \;:\equiv\; \exists^{\mathrm{st}} x^{(\tau^+\Phi)^\circ} \forall^{\mathrm{st}} y^{\tau^-\Phi} \, |\Phi|^x_y$

Soundness for the parametrised formula interpretation

Theorem. Let Δ_{int} be a set of internal formula. If

$$\mathsf{HA}^{\omega *}_{\mathsf{st}} + \Delta_{\mathrm{int}} \vdash \Phi$$

then from the proof we can extract a closed term $t : (\tau^{+}\Phi)^{\circ}$ in T° ($:\equiv \mathsf{T}^{*} + \circ$) such that

$$\mathsf{HA}^{\circ} + \Delta_{\mathrm{int}} \vdash \forall y^{\mathsf{d}^{-}\Phi} |\Phi|^{t}_{y}.$$

Proof. By induction on the length of the derivation.

Instantiations of the parametrised formula interpretation

σ°	$t\ \epsilon\ u$	$\forall x \sqsubset t\, A(x)$	Functional interpretations
σ	$t = u$	$A(t)$	Dialectica interpretation
σ	$t = u$	$\forall^{\mathrm{st}} x A(x)$	Modified realisability
σ	$t \leq^* u$	$\tilde{\forall} x \leq^* t\, A(x)$	Bounded functional interpretation[4]
			$\vdots$
σ^*	$t \in u$	$\forall x \in t\, A(x)$	Herbrand Dialectica interpretation (Diller-Nahm)
σ^*	$t \in u$	$\forall^{\mathrm{st}} x A(x)$	Herbrand realisability
			$\vdots$

- One interpretation of "standardness" is totality.
- Then $\forall^{\mathrm{st}}$, $\exists^{\mathrm{st}}$ are the computational quantifiers in Berger's uniform HA.

[4] Fernando Ferreira, Jaime Gaspar, *Nonstandardness and the bounded functional interpretation*, Annals of Pure and Applied Logic 166 (2015), no. 6, 701–712.

Summary

Work done:

- We reformulate Herbrand functional interpretations in a way that is suitable for a type-theoretic development.
- Then we embed H within the Agda proof assistant and implement the term extraction algorithm via the Herbrand Dialectica interpretation.
- We extend Oliva's method to unify (Herbrand) functional interpretations of (nonstandard) arithmetic.

Work to do:

- Add more examples (incl. real numbers) to the implementation.
- Complete the formalisation of the soundness proof.
- Adapt Schwichtenberg and Ratiu's decoration algorithm to the parametrised formula interpretation.